

DATA 583

Advanced Predictive Modelling



Chapter 4: Another Look at Multiple Regression

- **Developing multiple regression models**
- **Least-squares estimation via the QR decomposition**
- **Inference: Estimation and Prediction**
- **F -tests**

Developing multiple regression models - Example

The data frame `table.b4` in the *MPV* package contains the following columns:

```
y sale price of the house (in thousands of dollars)
x1 taxes (in thousands of dollars)
x2 number of baths
x3 lot size (in thousands of square feet)
x4 living space (in thousands of square feet)
x5 number of garage stalls
x6 number of rooms
x7 number of bedrooms
x8 age of the home (in years)
x9 number of fireplaces
```

There are 24 observations on these variables in the data frame.

Can we use predict the sale price of a house based on the other information?

The sale price is an example of a response variable, and the covariates which are used in a model to predict or explain sale price are referred to as predictor variables or explanatory variables.

Developing multiple regression models

Recall the simple linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

where the noise or error, $\varepsilon_i \sim N(0, \sigma^2)$ (independent).

The normality assumption could be relaxed.

We extend this model to a multiple regression model by adding more variables on the right side, that is, more explanatory variables.

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i$$

For the house sale price example, if we included all covariates in the model, we would have $k = 9$ terms in the model in addition to the intercept.

Developing multiple regression models

The multiple regression framework includes special types of nonlinear models.

For example, we could try to model the response as a k th degree polynomial.

Such a polynomial regression model is given by

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_k x_i^k + \varepsilon_i$$

We could also include polynomial terms in several different variables, such as x_1^2 and x_2^2 as well as interaction terms, such as $x_1 x_2$ or $x_1^2 x_2$. Therefore, being able to construct multiple regression models gives a great deal of flexibility to how data can be analyzed.

Matrix form

Handling multiple covariates is more convenient if we use vectors and matrices.

The multiple regression model can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad (1)$$

where $\mathbf{y}$ is the column vector of responses:

$$\mathbf{y} = [y_1 \ y_2 \ \cdots \ y_n]^\top.$$

On the right hand side of (1), we have

$$\boldsymbol{\beta} = [\beta_0 \ \beta_1 \ \cdots \ \beta_k]^\top$$

and

$$\boldsymbol{\epsilon} = [\epsilon_1 \ \epsilon_2 \ \cdots \ \epsilon_n]^\top.$$

Matrix form

X is an $n \times (k + 1)$ matrix, called the **model matrix** or **design matrix**:

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}$$

Matrix form

The following special cases can help to illustrate the notation.

Regression through the origin:

$$y_i = \beta_1 x_i + \epsilon_i$$

is

$$\mathbf{y} = \mathbf{X}\beta + \boldsymbol{\epsilon}$$

with

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \quad \text{and} \quad \beta = [\beta_1]$$

Matrix form

Simple linear regression:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

is

$$\mathbf{y} = \mathbf{X}\beta + \epsilon$$

with

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \quad \text{and} \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

Matrix form

Quadratic regression:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i$$

is

$$\mathbf{y} = \mathbf{X}\beta + \boldsymbol{\epsilon}$$

with

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \quad \text{and} \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

Matrix form

Regression with 2 predictor variables

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i$$

is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

with

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \quad \text{and} \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

The QR decomposition of a matrix

For any $n \times p$ matrix X whose columns are linearly independent, there exists an $n \times n$ orthogonal matrix Q and an invertible $n \times p$ upper triangular matrix R , such that

$$X = QR.$$

Orthogonality of Q means that $Q^T Q = I$.

In order for the columns of X to be linearly independent, it is necessary that $p \leq n$.

For our ultimate purpose, we will need to assume that $p < n$, in fact.

Computational verification of the QR decomposition

Here, we outline the steps to verify that a given design matrix X has a QR decomposition. For example, suppose

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 5 \\ 1 & 0 & 3 \\ 1 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}.$$

The matrices Q and R can be obtained as follows:

```
p <- 3
X <- matrix(c(rep(1, 5),
               2, 1, 0, 1, 2,
               3, 5, 3, 1, 0),
            ncol=p)
X.QR <- qr(X)
Q <- qr.Q(X.QR, complete=TRUE)
R <- qr.R(X.QR, complete=TRUE)
```

Computational verification of the QR decomposition

We can verify that $X = QR$ by computing the difference and rounding to 14 decimal places.

```
round(X - Q%*%R, digits = 14)  # Subtracting QR from X
```

```
##      [,1] [,2] [,3]
## [1,]    0    0    0
## [2,]    0    0    0
## [3,]    0    0    0
## [4,]    0    0    0
## [5,]    0    0    0
```

Computational verification of the QR decomposition

The orthogonality of Q can be verified by comparing $Q^T Q$ with the identity matrix:

```
I <- diag(rep(1, 5))
round(I - t(Q) %*% Q, 14)
```

	[, 1]	[, 2]	[, 3]	[, 4]	[, 5]
[1,]	0	0	0	0	0
[2,]	0	0	0	0	0
[3,]	0	0	0	0	0
[4,]	0	0	0	0	0
[5,]	0	0	0	0	0

Computational verification of the QR decomposition

The matrix R is upper triangular:

```
R

##           [, 1]      [, 2]      [, 3]
## [1, ] -2.236068 -2.683282 -5.366563
## [2, ]  0.000000  1.673320 -1.434274
## [3, ]  0.000000  0.000000  3.625308
## [4, ]  0.000000  0.000000  0.000000
## [5, ]  0.000000  0.000000  0.000000
```

Note that only the top p rows of R are nonzero:

$$R = \begin{bmatrix} U \\ 0 \end{bmatrix}$$

where U is a $p \times p$ upper triangular matrix.

A useful partition of Q

Partition the matrix Q as follows.

$$Q = [Q_1 \ Q_2]$$

where Q_1 consists of the first p columns of Q .

Verify the following:

$$Q_1^T Q_1 = \mathbf{I}$$

$$Q_1^T Q_2 = 0$$

$$Q_2^T Q_1 = 0$$

and

$$Q_2^T Q_2 = \mathbf{I}.$$

Also,

$$X = QR = Q_1 U.$$

Numerical verification of the properties of partitioned Q

```
Q1 <- Q[, 1:p]
Q2 <- Q[, -(1:p)]
round(t(Q1) %*% Q1, 14)
```

```
##      [, 1] [, 2] [, 3]
## [1, ]    1    0    0
## [2, ]    0    1    0
## [3, ]    0    0    1
```

```
round(t(Q1) %*% Q2, 14)
```

```
##      [, 1] [, 2]
## [1, ]    0    0
## [2, ]    0    0
## [3, ]    0    0
```

Numerical verification of the properties of partitioned Q

round(**t**(Q2) %*%Q1, 14)

##		[, 1]	[, 2]	[, 3]
##	[1,]	0	0	0
##	[2,]	0	0	0

round(**t**(Q2) %*%Q2, 14)

##		[, 1]	[, 2]
##	[1,]	1	0
##	[2,]	0	1

Numerical verification of the properties of partitioned Q

```
U <- R[1:p, ]
round(Q1 %*% U, 14)
```

```
##      [,1] [,2] [,3]
## [1,]    1    2    3
## [2,]    1    1    5
## [3,]    1    0    3
## [4,]    1    1    1
## [5,]    1    2    0
```

Observe that the diagonal elements of U are always nonzero when the columns of X are linearly independent.

Least-squares via the QR decomposition

The least-squares approach to estimating β in the regression model $\mathbf{y} = \mathbf{X}\beta + \epsilon$ is to minimize the **Residual Sum of Squares**

$$(\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$$

with respect to β . This can be done by using calculus, but it is better to note that, because $\mathbf{X} = \mathbf{Q}\mathbf{R}$, we have

$$\begin{aligned} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) &= (\mathbf{y} - \mathbf{Q}\mathbf{R}\beta)^\top (\mathbf{y} - \mathbf{Q}\mathbf{R}\beta) \\ &= (\mathbf{y} - \mathbf{Q}\mathbf{R}\beta)^\top (\mathbf{Q}\mathbf{Q}^\top \mathbf{y} - \mathbf{Q}\mathbf{R}\beta) \\ &= (\mathbf{y} - \mathbf{Q}\mathbf{R}\beta)^\top \mathbf{Q}(\mathbf{Q}^\top \mathbf{y} - \mathbf{R}\beta) \\ &= (\mathbf{Q}^\top \mathbf{y} - \mathbf{R}\beta)^\top \mathbf{Q}^\top \mathbf{Q}(\mathbf{Q}^\top \mathbf{y} - \mathbf{R}\beta) \\ &= (\mathbf{Q}^\top \mathbf{y} - \mathbf{R}\beta)^\top (\mathbf{Q}^\top \mathbf{y} - \mathbf{R}\beta) \\ &= (\mathbf{Q}_1^\top \mathbf{y} - \mathbf{U}\beta)^\top (\mathbf{Q}_1^\top \mathbf{y} - \mathbf{U}\beta) + (\mathbf{Q}_2^\top \mathbf{y})^\top (\mathbf{Q}_2^\top \mathbf{y}). \end{aligned}$$

The term on the far right of this expression does not depend on β , so the minimizer of the entire expression is

$$\hat{\beta} = \mathbf{U}^{-1} \mathbf{Q}_1^\top \mathbf{y}.$$

Parameter estimation via QR

The QR approach is worth reiterating for clarity.

Model: $\mathbf{y} = \mathbf{X}\beta + \epsilon = \mathbf{Q}\mathbf{R}\beta + \epsilon = \mathbf{Q}_1\mathbf{U}\beta + \epsilon$ where we have partitioned $\mathbf{Q}$ as $[\mathbf{Q}_1 \ \mathbf{Q}_2]$. $\mathbf{Q}_1$ represents the first p columns of $\mathbf{Q}$.

Multiplying the model equation on the left by $\mathbf{Q}^\top$ gives

$$\mathbf{Q}^\top \mathbf{y} = \mathbf{R}\beta + \mathbf{Q}^\top \epsilon$$

which can be re-written as

$$\mathbf{Q}_1^\top \mathbf{y} = \mathbf{U}\beta + \mathbf{Q}_1^\top \epsilon \quad (2)$$

and

$$\mathbf{Q}_2^\top \mathbf{y} = \mathbf{Q}_2^\top \epsilon \quad (3)$$

Parameter estimation via QR

To estimate β , it is clear that model (2) contains all of the needed information, and the ordinary least-squares estimate must be the solution of the upper triangular linear system

$$U\hat{\beta} = Q_1^T y$$

since the solution of this must minimize the least-squares objective function

$$(Q_1^T y - U\beta)^T (Q_1^T y - U\beta)$$

Parameter Estimation via QR

To estimate σ^2 , model equation (3) is used. It is easily shown that

$$E[Q_2^T \mathbf{y} | \mathbf{X}] = 0$$

and

$$\text{Var}(Q_2^T \mathbf{y} | \mathbf{X}) = Q_2^T \sigma^2 \mathbf{I} Q_2 = \sigma^2 \mathbf{I}.$$

Thus, $Q_2^T \mathbf{y}$ is a vector of $n - p$ uncorrelated mean 0 and variance σ^2 random variables.

Hence, $\mathbf{y}^T Q_2 Q_2^T \mathbf{y}$ must be the sum of squares of such random variables.

$$\text{i.e.} \quad \mathbf{y}^T Q_2 Q_2^T \mathbf{y} = \sum_{i=1}^{n-p} Z_i^2 \quad (4)$$

where $Z_1, Z_2, \dots, Z_{n-p}$ are uncorrelated random variables with mean 0 and variance $\text{Var}(Z_i) = \sigma^2$.

Parameter Estimation via QR

Therefore,

$$E[\mathbf{y}^\top \mathbf{Q}_2 \mathbf{Q}_2^\top \mathbf{y}] = \sum_{i=1}^{n-p} \text{Var}(Z_i) = (n-p)\sigma^2,$$

⇒ An unbiased estimator for σ^2 is

$$\hat{\sigma}^2 = \text{MSE} = \frac{\mathbf{y}^\top \mathbf{Q}_2 \mathbf{Q}_2^\top \mathbf{y}}{n-p}$$

Note that the residual sum of squares is $\text{SSE} = \mathbf{y}^\top \mathbf{Q}_2 \mathbf{Q}_2^\top \mathbf{y}$.

If we make the further assumption that $\mathbf{y}$ is normal, then the discussion around (4) implies that the SSE is a sum of squares of uncorrelated normal random variables Z_i with mean 0 and variance $\text{Var}(Z_i) = \sigma^2$.

Therefore, in the normal case, SSE/σ^2 is a χ^2 random variable on $n-p$ degrees of freedom.

Also, $Q_2^T Q_1 = 0$. $Q_1^T y$ and $Q_2^T y$ are vectors of uncorrelated normal random variables, hence, they are independent.

Therefore, $\hat{\beta}$ (which depends only on $Q_1^T y$) and **MSE** must be independent, when the data are normally distributed.

Parameter Estimation via QR - Example

We illustrate our techniques on the house sales data in `table.b4`.

```
X <- table.b4[, -1] # remove "y" column to form model matrix
n <- nrow(table.b4); X1 <- rep(1, n) # column for intercept
X <- cbind(X1, X) # append 1's to create full model matrix
QR <- qr(X) # QR decomposition of X
Q <- qr.Q(QR, complete=TRUE) # complete=TRUE gives Q1 & Q2
p <- 10 # p = 9+1 for this problem
Q1 <- Q[, 1:p]
y <- table.b4[, 1]
Q1y <- t(Q1) %*% y
U <- qr.R(QR) # if you want all of R, use complete=TRUE
betahat <- solve(U, Q1y)
```

Parameter Estimation via QR - Example

Parameter Estimates:

```
betahat
```

```
##           [,1]
## x1 14.92764759
## x1  1.92472156
## x2  7.00053420
## x3  0.14917793
## x4  2.72280790
## x5  2.00668402
## x6 -0.41012376
## x7 -1.40323530
## x8 -0.03714908
## x9  1.55944663
```

Fitted model:

$$\widehat{E[y|x]} = 14.9 + 1.9x_1 + 7x_2 + 0.1x_3 + 2.7x_4 + 2x_5 - 0.4x_6 - 1.4x_7 - 0.04x_8 + 1.6x_9$$

This can also be obtained from

```
y.lm <- lm(y ~ ., data = table.b4)
coef(y.lm)
```

Parameter Estimation via QR - Example

Residual Sum of Squares:

```
Q2 <- Q[, -(1:p)]
Q2y <- t(Q2) %*% y
SSE <- t(Q2y) %*% Q2y
SSE

##           [,1]
## [1,] 121.7482
```

Variance Estimate and Root-MSE*:

```
MSE <- SSE / (n-p)
MSE

##           [,1]
## [1,] 8.696297
```

```
sqrt(MSE)

##           [,1]
## [1,] 2.948949
```

The root-MSE is part of the output from `summary(y.lm)`.

*This is the estimate of σ , the noise standard deviation.

Properties of the Parameter Estimates

Unbiasedness of $\hat{\beta} = U^{-1}Q_1^T y$ is easily verified, since it can be shown that

$$\hat{\beta} = \beta + U^{-1}Q_1^T \epsilon.$$

This last result can also be used to see that

$$\text{Var}(\hat{\beta}) = U^{-1}Q_1^T \text{Var}(\epsilon)Q_1 U^{-T} = \sigma^2 U^{-1}U^{-T}$$

What have we assumed?

Not very much:

- the noise terms are uncorrelated with each other and with the covariates.
- the noise variance is constant.*

*This is needed to assess uncertainty in the model.

Estimating the mean response at $\mathbf{x}_0$

The expected value of y at given values of the explanatory variables, summarized by the row vector $\mathbf{x}_0^*$ is given by

$$E[y|\mathbf{x}_0] = \mathbf{x}_0\beta.$$

We estimate this with

$$\widehat{E[y|\mathbf{x}_0]} = \mathbf{x}_0\hat{\beta}.$$

*which is of length p and whose first component must be '1', if an intercept has been included in the model

Estimating the mean response at x_0 - Example

Estimate the mean sale price for a house with characteristics summarized in a 9-vector called x_0 :

```
##      x1      x2      x3      x4      x5      x6      x7      x8      x9
##    4.5    1.0    2.3    1.2    1.0    6.0    3.0   40.0    0.0
```

```
x0 <- c(1, x0)
Ey.hat <- x0 %*% betahat  # estimated price in thousands
```

Estimated mean sale price:

```
1000 * Ey.hat

##              [,1]
## [1,] 28050.18
```

Estimation error

We can assess the uncertainty in this estimate by calculating the standard error of this estimate.

The variance of the estimator is

$$\text{Var}(\mathbf{x}_0 \hat{\beta}) = \mathbf{x}_0 \text{Var}(\hat{\beta}) \mathbf{x}_0^T = \sigma^2 \mathbf{x}_0 (\mathbf{U}^{-1} \mathbf{U}^{-T}) \mathbf{x}_0^T.$$

The standard error of the mean response at $\mathbf{x}_0$ is, therefore,

$$\sigma \sqrt{\mathbf{x}_0 (\mathbf{U}^{-1} \mathbf{U}^{-T}) \mathbf{x}_0^T}.$$

Replacing σ by $\hat{\sigma}$ in this formula gives us a practical estimator for the standard error.

Estimation error - Example

For the house price data, we have estimated the mean response at x_0 to be 28.0501799. The standard error of this estimate is

```
UTx0T <- solve(t(U), x0)
SE <- sqrt(MSE) * sqrt(sum(UTx0T^2))
SE

##           [,1]
## [1,] 1.48408
```

We can multiply this value by \$1000 to get the standard error in actual dollars:

```
1000 * SE

##           [,1]
## [1,] 1484.08
```

Confidence interval for the mean response

If y is normally distributed, $\frac{\mathbf{x}_0\hat{\beta} - \mathbf{x}_0\beta}{\sigma\sqrt{\mathbf{x}_0(\mathbf{U}^{-1}\mathbf{U}^{-\top})\mathbf{x}_0^{\top}}}$ must be standard normal, and because SSE/σ^2 is distributed as $\chi^2_{(n-p)}$, $\frac{\mathbf{x}_0\hat{\beta} - \mathbf{x}_0\beta}{\sqrt{\text{MSE}}\sqrt{\mathbf{x}_0(\mathbf{U}^{-1}\mathbf{U}^{-\top})\mathbf{x}_0^{\top}}}$ must have a t distribution on $n - p$ degrees of freedom, since $\hat{\beta}$ and $\text{MSE} = \text{SSE}/(n - p)$ are independent of each other.

A $1 - \alpha$ confidence interval for $E[y|\mathbf{x}_0] = \mathbf{x}_0\beta$ is then

$$\mathbf{x}_0\hat{\beta} \pm t_{n-p, \alpha/2} \sqrt{\text{MSE}} \sqrt{\mathbf{x}_0(\mathbf{U}^{-1}\mathbf{U}^{-\top})\mathbf{x}_0^{\top}}.$$

Confidence interval for the mean response - Example

A 95% confidence interval for the mean sale price of the house at x_0 (in thousands) is

```
CI <- c(Ey.hat - qt(.975, df = n - p)*SE, Ey.hat +
        qt(.975, df = n - p)*SE)
```

```
CI
```

```
## [1] 24.86714 31.23321
```

In dollars, this is

```
1000*CI
```

```
## [1] 24867.14 31233.21
```

Prediction error

We may also wish to predict a single response value y for given values of the explanatory variables, summarized by the row vector $\mathbf{x}_0^*$.

- use the estimated response for the point prediction
- take into account not only error in the estimate of the mean response, but also for the prediction error associated with any variables or factors that have not been measured (i.e. the noise ε in the new response).

The variance of the prediction error is

$$\text{Var}(\mathbf{x}_0 \hat{\beta} + \varepsilon) = \underbrace{\mathbf{x}_0 (\mathbf{U}^{-1} \mathbf{U}^{-\top}) \mathbf{x}_0^{\top} \sigma^2}_{\text{error in mean estimate}} + \underbrace{\sigma^2}_{\text{noise in new response}}.$$

The standard deviation of the prediction error is thus

$$\sigma \sqrt{\mathbf{x}_0 (\mathbf{U}^{-1} \mathbf{U}^{-\top}) \mathbf{x}_0^{\top} + 1}.$$

*which is of length p and whose first component must be '1', if an intercept has been included in the model

Prediction error - Example

For the house price data, we have estimated the mean response at x_0 to be 28.0501799. For a single house price, the standard error of this estimate as a prediction is

```
UTx0T <- solve(t(U), x0)
SEPred<- sqrt(MSE)*sqrt(1+ sum(UTx0T^2))
SEPred

##           [,1]
## [1,] 3.301332
```

We can multiply this value by \$1000 to get the standard error of prediction in actual dollars:

```
1000*SEPred

##           [,1]
## [1,] 3301.332
```

Prediction interval for a new response

If ϵ is normally distributed, $\frac{\mathbf{x}_0\hat{\beta} + \epsilon - \mathbf{x}_0\beta}{\sigma\sqrt{1 + \mathbf{x}_0(\mathbf{U}^{-1}\mathbf{U}^{-\top})\mathbf{x}_0^\top}}$ must be standard normal, and because SSE/σ^2 is distributed as $\chi^2_{(n-p)}$, $\frac{\mathbf{x}_0\hat{\beta} + \epsilon - \mathbf{x}_0\beta}{\sqrt{\text{MSE}}\sqrt{1 + \mathbf{x}_0(\mathbf{U}^{-1}\mathbf{U}^{-\top})\mathbf{x}_0^\top}}$ must have a t distribution on $n-p$ degrees of freedom, since $\hat{\beta}$ and $\text{MSE} = \text{SSE}/(n-p)$ are independent of each other.

A $1 - \alpha$ prediction interval for $Y = \mathbf{x}_0\beta + \epsilon$ is then

$$\mathbf{x}_0\hat{\beta} \pm t_{n-p, \alpha/2} \sqrt{\text{MSE}} \sqrt{1 + \mathbf{x}_0(\mathbf{U}^{-1}\mathbf{U}^{-\top})\mathbf{x}_0^\top}.$$

This interval should contain the new response $Y = \mathbf{x}_0\beta + \epsilon$ with probability $1 - \alpha$.

Prediction interval for a new response - example

A 95% prediction interval for the house price for a single house having the characteristics described in x_0 is

```
PI <- c(Ey.hat - qt(.975, df = n - p)*SEPred, Ey.hat +  
        qt(.975, df = n - p)*SEPred)
```

```
PI
```

```
## [1] 20.96953 35.13083
```

In dollars, this is

```
1000*PI
```

```
## [1] 20969.53 35130.83
```

Inference for the regression coefficients

It can also be deduced that $\hat{\beta}$ is normally distributed with mean β and variance-covariance matrix $(U^{-1}U^{-\top})\sigma^2$. Thus, each component of $\hat{\beta}$ can be standardized as follows:

$$\frac{\hat{\beta}_i - \beta_i}{\sigma \sqrt{(U^{-1}U^{-\top})_{i+1,i+1}}}$$

to become a standard normal random variable, for $i = 0, 1, 2, \dots, p - 1$.

It then follows, by the same reasoning as before, that

$$\frac{\hat{\beta}_i - \beta_i}{\hat{\sigma} \sqrt{(U^{-1}U^{-\top})_{i+1,i+1}}}$$

must have a t -distribution on $n - p$ degrees of freedom.

This result can be used to justify t -tests and related confidence intervals for individual regression coefficients.

A confidence interval for a regression coefficient

Let c_{jj} be the $j + 1$ th diagonal element of $(\mathbf{U}^\top \mathbf{U})^{-1}$. Then a $(1 - \alpha)$ confidence interval for β_j is

$$\hat{\beta}_j \pm t_{n-p, \alpha/2} \hat{\sigma} \sqrt{c_{jj}} = \hat{\beta}_j \pm t_{n-p, \alpha/2} SE_j$$

where we have denoted the Standard Error of the j th coefficient estimate as SE_j .

A confidence interval for a regression coefficient - example

For the house price data, the coefficient standard errors can be computed using

```
Cii <- sqrt(diag(solve(t(U) %*% U)))
SEii <- Cii * as.numeric(sqrt(MSE))
SEii
```

	X1	x1	x2	x3	x4	x5
##	5.91285	1.02990	4.30037	0.49039	4.35955	1.37351
	x6	x7	x8	x9		
##	2.37854	3.39554	0.06672	1.93750		

Compare with the standard errors obtained from `summary(y.lm)`.

A 95% confidence interval for the 4th regression coefficient (β_3) is

```
betahat[4] + SEii[4] * qt(c(.025, .975), df = n-p)
```

```
## [1] -0.9026 1.2010
```

F statistics

Given $r \in \{1, \dots, p\}$, consider the (null) hypothesis that

$$\beta_{p-r} = \beta_{p-r+1} = \dots = \beta_{p-1} = 0$$

From the QR decomposition of X and model (2), we have, under the above hypothesis,

$$\begin{aligned}
 E[\mathbf{Q}_1^T \mathbf{y}] &= \begin{bmatrix} U_{1,1}\beta_0 + U_{1,2}\beta_1 + \dots + U_{1,p}\beta_{p-1} \\ U_{2,2}\beta_1 + \dots + U_{2,p}\beta_{p-1} \\ \dots \\ \dots \\ \dots \\ \dots \\ U_{p,p}\beta_{p-1} \end{bmatrix} \\
 &= \begin{bmatrix} U_{1,1}\beta_0 + U_{1,2}\beta_1 + \dots + U_{1,p-r}\beta_{p-r-1} \\ U_{2,2}\beta_1 + \dots + U_{2,p-r}\beta_{p-r-1} \\ \dots \\ U_{p-r,p-r}\beta_{p-r-1} \\ 0 \\ \dots \\ 0 \end{bmatrix}
 \end{aligned}$$

F statistics

Partition the vector $Q_1^T y$ as

$$Q_1^T y = \begin{bmatrix} q_{11} \\ q_{12} \end{bmatrix}$$

where q_{12} denotes the last r components of $Q_1^T y$

q_{12} is a vector of r independent normal random variables with mean 0 and variance σ^2 , when the null hypothesis is true.

F statistics

Therefore,

$$\frac{q_{12}^T q_{12}}{\sigma^2}$$

must have a χ^2 distribution on r degrees of freedom, and because $Q_1^T Q_2 = 0$, q_{12} is independent of $\hat{\sigma}^2$, so

$$\frac{q_{12}^T q_{12}/r}{\hat{\sigma}^2}$$

must have an F distribution on r and $n - p$ degrees of freedom, when the null hypothesis is true.

F statistics - Example

Let's test whether the last 4 regression coefficients in the house price data set are 0 versus the alternative that at least one is nonzero. This means that $r = 4$, and q_{12} takes elements 7 through 10, of $Q_1^T y$.

```
q12 <- Q1y[7:10]
Fstat.numerator <- sum(q12^2)/4
Fstat <- Fstat.numerator/MSE
Fstat

##           [,1]
## [1,] 0.8179

1 - pf(Fstat, 4, n - p) # p-value for this test

##           [,1]
## [1,] 0.5348
```

The p-value is very large. We fail to reject the null hypothesis. There is no evidence that any of the last 4 regression coefficients are nonzero.

This calculation can also be done in R with the `anova` function:

```
y0.lm <- lm(y ~ x1 + x2 + x3 + x4 + x5, data = table.b4)
anova(y0.lm, y.lm)
```

F statistics

The F -test for significance of regression corresponds to the case where $r = p - 1$

Then $E[q_{11}] = U_{1,1}\beta_0$.

$E[q_{12}]$ is a $(p - 1)$ -vector of 0's under the null hypothesis and must have at least one nonzero component if the alternative hypothesis is true.

F statistics

This is a test for whether **any** of the regression coefficients in the house price data set are 0 versus the alternative that at least one is nonzero. This means that $r = 9$, and q_{12} takes elements 2 through 10, of $Q_1^T y$. (We don't bother testing the intercept.)

```
q12 <- Q1y[2:10]
Fstat.numerator <- sum(q12^2) / 9
Fstat <- Fstat.numerator / MSE
Fstat

##           [,1]
## [1,] 9.037

1 - pf(Fstat, 9, n - p) # p-value for this test

##           [,1]
## [1,] 0.000185
```

The p-value is microscopic. We reject the null hypothesis. At least one of the regression coefficients is nonzero. We don't know which one(s).

This p-value is printed at the bottom of `summary(y.lm)` and can also be done in R with the `anova` function:

```
y0.lm <- lm(y ~ 1, data = table.b4) # model with intercept only
anova(y0.lm, y.lm)
```